

ECON 5345 Homework 11 Report

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Starting from the initial guess $V_0(a) = \vec{0}$, VFI result is shown below in Figure 1.

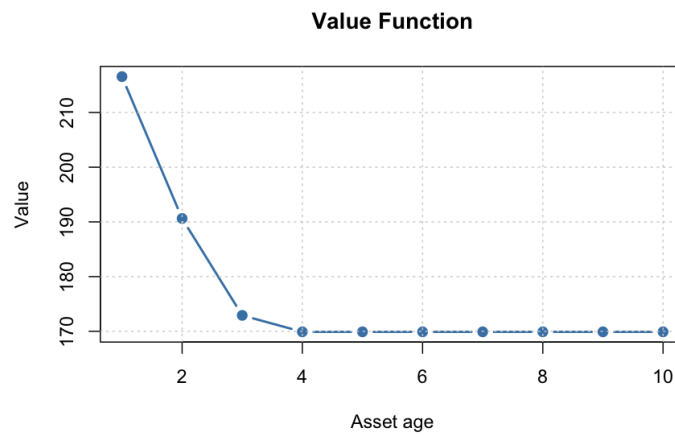


Figure 1: Value function

We can see that the optimal replacement policy is to replace the asset when it is 4 years old. Hence, the path of state variable is in Figure 2.

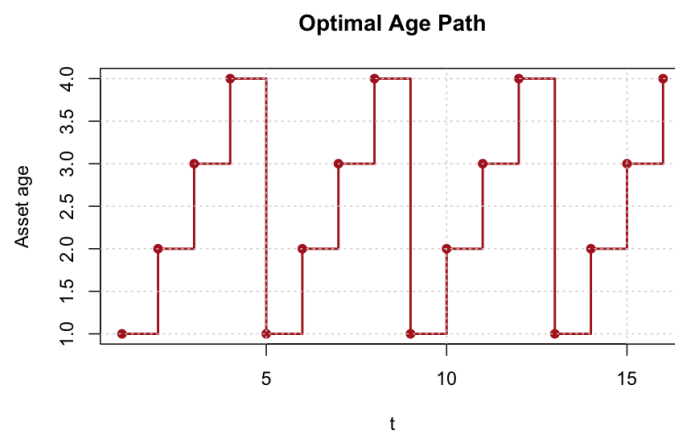


Figure 2: Optimal age path

To prove that 4 is really the cutoff age, we can solve the problem analytically. Guess that the cutoff age is $a^* > 1$. Then

$$V(a) \equiv V^* := p(0) - c + \beta V(1) = 0.9V(1) - 25, \forall a \geq a^*.$$

Note that

$$V(a) = \sum_{j=0}^{a^*-a-1} p(a+j) + \beta^{a^*-a} V^*$$

To verify that $a^* = 4$, what we need to show is that $p(4) + 0.9V^* \leq V^*$ and $p(3) + 0.9V^* > V^*$. Note that $p(1) = 45$, $p(2) = 35$, $p(3) = 20$, $p(4) = 0$, so we have

$$p(4) + 0.9V^* = 0 + 0.9V^* \leq V^*.$$

By backward induction formula, we have $V(1) = 92.7 + 0.729V^*$. Substituting this into $V^* = 0.9V(1) - 25$, we get $V^* = 169.9040$. Hence $20 + 0.9V^* > V^*$.

Therefore, $a^* = 4$ is actually the cutoff.